

1.Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
-------	--------

9	
---	--

12	
----	--

Answer:(penalty regime: 0 %)

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n;
```

```
    scanf("%d", &n);
```

```
    int count = 0;
```

```
    int i = 1;
```

```
    count++;
```

```
    int s = 1;
```

```
    count++;
```

```
    while (1)
```

```
    {
```

```
        count++;
```

```
        if (!(s <= n))
```

```
            break;
```

```
        i++;
```

```
        count++;
```

```
        s += i;
```

```
        count++;
```

```
    }
```

```
    printf("%d", count);
```

```
    return 0;  
}
```

Feedback

Input	Expected	Got
9		
12		
12		
4		
9		
9		

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

2.Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)  
{  
    if(n==1)  
    {
```

```

    printf("*");
}
else
{
    for(int i=1; i<=n; i++)
    {
        for(int j=1; j<=n; j++)
        {
            printf("*");
            printf("*");
            break;
        }
    }
}
}

```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:(penalty regime: 0 %)

#include <stdio.h>

```

int main()
{
    int n;

    scanf("%d", &n);


    int count;


    if (n == 1)
    {
        count = 1;
    }
    else
    {
        count = 5 * n + 2;
    }


    printf("%d", count);


    return 0;
}

```

	Input	Expected	Got
2			
12			
12			

1000

5002

5002

143

717

717

Passed all tests!

3.Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {  
    {  
        for (i = 1; i <= num;++i)  
        {  
            if (num % i== 0)  
            {  
                printf("%d ", i);  
            }  
        }  
    }  
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int num;
```

```
    scanf("%d", &num);
```

```
    int count = 0;
```

```
    int i;
```

```
    for (i = 1; i <= num; i++)
```

```
    {
```

```
        count++;
```

```
        count++;
```

```
        if (num % i == 0)
```

```
        {
```

```
            count++;
```

```
        }
```

```
    }
```

```
    count++;
```

```
printf("%d", count);
```

```
return 0;
```

```
}
```

Input	Expected	Got
-------	----------	-----

12		
----	--	--

31		
----	--	--

31		
----	--	--

25		
----	--	--

54		
----	--	--

54		
----	--	--

4		
---	--	--

12		
----	--	--

12		
----	--	--

Passed all tests!

4.Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
```

```
{
```

```
    int c= 0;
```

```
    for(int i=n/2; i<n; i++)
```

```
        for(int j=1; j<n; j = 2 * j)
```

```
            for(int k=1; k<n; k = k * 2)
```



```
        c++;  
    }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include <stdio.h>  
  
#include <math.h>  
  
int main(){  
  
    int n;  
  
    scanf("%d", &n);  
  
    int count = 1;  
  
    for (int i = n / 2; i < n; i++){  
  
        count++;  
  
        for (int j = 1; j < n; j = 2 * j){  
  
            count++;  
  
            for (int k = 1; k < n; k = k * 2){  
  
                count++;  
  
                count++;}  
  
            count++;}  
  
        count++;}  
  
    }
```

```

        count++;}

count++;

printf("%d", count);

return 0;

}

```

Input	Expected	Got
-------	----------	-----

4		
---	--	--

30		
----	--	--

30		
----	--	--

10		
----	--	--

212		
-----	--	--

212		
-----	--	--

4.Convert the following algorithm into a program and find its time complexity using counter method.

```

void function(int n)
{
    int c= 0;

    for(int i=n/2; i<n; i++)

        for(int j=1; j<n; j = 2 * j)

            for(int k=1; k<n; k = k * 2)

                c++;
}

```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include <stdio.h>
```

```
#include <math.h>
```

```
int main(){
```

```
    int n;
```

```
    scanf("%d", &n);
```

```
    int count = 1;
```

```
    for (int i = n / 2; i < n; i++){
```

```
        count++;
```

```
        for (int j = 1; j < n; j = 2 * j){
```

```
            count++;
```

```
            for (int k = 1; k < n; k = k * 2){
```

```
                count++;
```

```
                count++;}
```

```
            count++;}
```

```
        count++;}
```

```
    count++;
```

```
    printf("%d", count);
```

```
    return 0;
```

```
}
```

Input	Expected	Got
-------	----------	-----

4		
---	--	--

30		
----	--	--

30		
----	--	--

10		
----	--	--

212		
-----	--	--

212		
-----	--	--

Passed all tests!

5.Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
```

```
{
```

```
    int rev = 0, remainder;
```

```
    while (n != 0)
```

```
    {
```

```
        remainder = n % 10;
```

```
        rev = rev * 10 + remainder;
```

```
        n/= 10;
```

```
    }
```

```
}
```

```
print(rev);
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

```
#include <stdio.h>
```

```
int main(){
```

```
    int n;
```

```
    scanf("%d", &n);
```

```
    int temp = n;
```

```
    int digits = 0;
```

```
    while (temp != 0){
```

```
        digits++;
```

```
        temp /= 10;
```

```
    }
```

```
    int count = 4 * digits + 3;
```

```
    printf("%d", count);
```

```
    return 0;
```

```
}
```

Input	Expected	Got
-------	----------	-----

12		
----	--	--

11		
----	--	--

11		
----	--	--

1234		
------	--	--

19		
----	--	--

19

Passed all tests!